

PROJECT 1: PROBLEM STATEMENT

Smart Product Recommendation System using Collaborative Filtering

Problem Statement

Users face **too many choices** on platforms like e-commerce and streaming services

Difficult to find **relevant products or content**

Traditional search methods are **not personalized**

Leads to **information overload and poor user experience**

Need for an **intelligent system** to analyze user behavior

System should use **past data (ratings, preferences)**

Goal: Provide **personalized recommendations**

Solution: Use **Collaborative Filtering technique**

Key Goals

- Analyze user-item interaction data
- Identify similar users or items
- Predict ratings for unseen items
- Recommend top relevant items to users

Real-World Applications

- Amazon (Product Recommendations)
- Netflix (Movie Recommendations)
- Spotify (Music Suggestions)
- YouTube (Video Recommendations)

Problem Challenges

- Cold start problem (new users/items)
- Data sparsity

- Scalability with large datasets
- Accuracy vs performance trade-off